

Assignment-II

Very Short Questions:

1. What do you mean by the complete binary tree and strictly binary tree?
2. Define Hash collision and collision resolution.
3. What do you mean by adaptive and non-adaptive sorting?
4. How many edges are there in a complete graph having 10 vertices?
5. Where can we use the Divide and Conquer algorithm? Write any two examples?
6. What are the differences between Kruskal's and Prim's algorithm that finds MST?
7. Explain BFS and DFS.
8. Why is external sorting slower than internal sorting?
9. How is B-tree different to Binary Search Tree?
10. Define the best case, worst case and average case of an algorithm.
11. What is a greedy algorithm?
12. What do you mean by Big Oh Notation?
13. What is a Binary Search Tree (BST)?
14. Define B tree with example.
15. Define Hash Function?
16. How does a merge sort algorithm work?
17. How does greedy algorithm work?
18. Why do you need minimum spanning tree?
19. What is complete binary tree?
20. What is hash function and hash table?
21. Define the leftist heap with leftist property.
22. What is a stable sort?
23. What is a directed graph?
24. What is a collision in a hash table?
- 25.** How does a priority queue differ from a regular queue?
26. How does a merge sort algorithm work?
27. Differentiate BFS and DFS.
28. Define dynamic programming.
- 29.** What is the lower bound for comparison based sorting algorithms?
30. Define Order and Shape property.
31. What is graph ?

Long Questions:

1. Define binary trees. Explain the following with suitable example. a. Complete binary tree b. Skewed Binary tree c. Full Binary Tree/Strict/Proper d. Almost Complete Binary Tree e. Degree of binary tree f. Height of binary tree
2. Differentiate between binary tree and binary search tree. What are the different binary tree traversal techniques? Explain.
3. Construct binary search tree from the following given string. Consider each character in the string represent information of node and also perform in-order, pre-order and post order traversal.

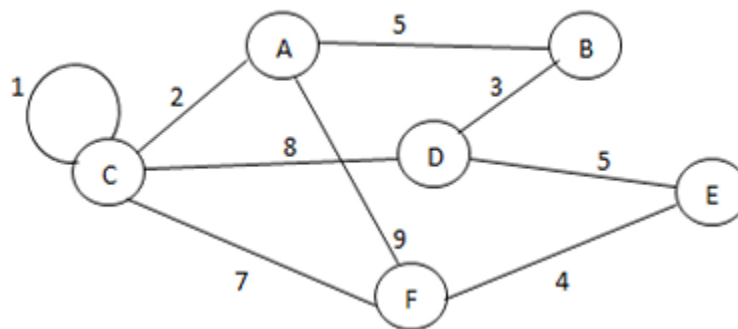
a. POKHARAUNIVERSITY

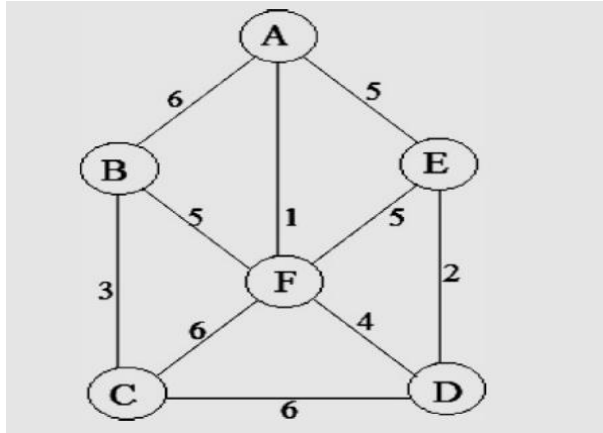
4. Reconstruct binary tree from following traversals.
 - a. In-Order Sequence: 4, 2, 5, 1, 6, 3, 7
 - b. Post-Order Sequence: 4, 5, 2, 6, 7, 3, 1 25.Reconstruct binary tree from following traversals.
 - a. In-Order Sequence: 2, 5, 6, 10, 12, 14, 15 b.
 - b. Post-Order Sequence: 10, 5, 2, 6, 14, 12, 15
5. Construct binary search tree a. 30, 40, 24, 58, 48, 26, 11, 13 b. 1,2,3,4,5,6,7
6. Define AVL Tree. Create an AVL Tree from the following data and show the deletion of the root. 30, 55, 65, 60, 50, 62, 20, 15, 12, 25
7. What do you mean by priority queue? Create a min -heap for the following data: 10, 15, 7, 3, 12, 20, 4, 8, 10, 30, 1
8. Sort Heap sort from the following data: 20, 10, 30, 25, 15, 40, 50, 35
9. Explain Quicksort in brief with an algorithm and suitable example.
10. How is a graph different from a tree? What is the basis of graph representation?
11. What is a collision in hashing? What are the collision-resolving techniques?
12. Discuss best-fit, worse-fit and average-fit analysis of the algorithm.
13. Sort the given data using Quick Sort:**54, 26, 93,17,77,31,44,55,20**
14. Explain binary search tree with example. Why do you need to balance a binary search tree? Explain in detail
15. What is sorting? Explain the bubble sort with example
16. Explain the Kruskal's algorithm with example.
17. Explain the concept of Randomized algorithms and Back Tracking algorithms with an example.
18. What is divide and conquer strategy? Explain in detail with the example of merge sort.
19. *Define minimum spanning tree. Write and explain Kruskal's algorithm to find minimum spanning tree with example.*
20. *Define backtracking algorithm. Write and explain one algorithm that uses this technique to solve a problem.*
21. *Sort the following numbers using bucket sort and insertion sort: 123, 456, 342, 789, 90, 2, 875, 453, 290, 998, 223*

22. Explain shell sort with its algorithm one suitable example.
23. Define graph and digraphs. Explain Adjacency matrix representation of graph with examples in undirected and directed graph.
24. Implement the simple hash function in C or Java code where $h(x) = x \% 10$.
25. Write a program for insertion sort.
26. What is asymptotic notation? Explain its type in detail. Which asymptotic notation is most frequently used and why?
27. Discuss the concept of best-case, worst -case and average case
28. Write a recursion algorithm to find Fibonacci Series upto N terms?

Case Questions :

1. Define Hash Function and Load Factor. Using the hash function $h(X) = X \bmod 5$, create a hash table for the following set of data: 11, 22, 37, 41, 53, 66, 72, 84
Construct the hash table using both linear probing and separate chaining to handle collisions.
2. Consider the given graph and construct a minimum spanning tree using Prim's algorithm and Kruskal's Algorithm.





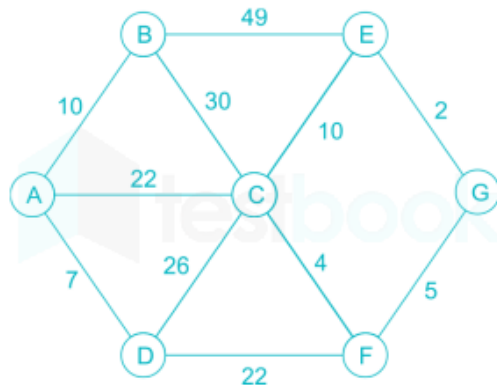
3. A queue is a container of objects (a linear collection) that are inserted and removed according to the first-in first-out (FIFO) principle. An excellent example of a queue is a line of students in the food court of the UC. New additions to a line made to the back of

the queue, while removal (or serving) happens in the front. In the queue only two operations are allowed enqueue and dequeue. Enqueue means to insert an item into the back of the queue, dequeue means removing the front item.

Question:

- Explain the operations performed in a queue
- Explain circular queue, double-ended queue and priority queue.
- Implement enqueue and dequeue

operations using the language C or Java for a circular queue.



- a) 4. Generate the Huffman code for the symbol A, B, C, D, E, F with the probability of occurrence is 2,8,3,6,1,4 respectively. Also construct Huffman tree.

5. Find shortest path algorithm Using Dijkstra's algorithm for the following graph.